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A Prototype Design of a Driver Fatigue Detection and Graded Response System Based on RISC-V

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***Abstract*—addressing the demand for low-cost, low-computing-power solutions in entry-level vehicles and the lack of engineering references for RISC-V in this domain, this paper designs and implements a prototype driver fatigue detection and graded response system based on a RISC-V MCU. The system adopts a layered and decoupled architecture: computationally intensive visual perception tasks are offloaded to a dedicated edge computing unit (K210), while the main control and decision-making are handled by a low-cost, automotive- grade 32-bit RISC-V MCU (EAM2011). This significantly reduces the computational burden on the main controller. By integrating classic visual fatigue features (EAR/PERCLOS/MAR) adapted for the edge unit and fusing them with steering wheel grip pressure signals using a "vision-primary, pressure-auxiliary" strategy, the system achieves a three-level graded response. Test results in a simulated driving environment show that under soft lighting conditions, The system attains a face-eye detection accuracy of at least ninety-two point five percent and an alarm latency of less than 300 milliseconds, and the system operates stably within an 8V-35V voltage range with basic protection. This work provides a modular and replicable reference solution for the engineering application of RISC-V in low-cost automotive safety systems.**

***Index Terms*—RISC-V architecture; driver fatigue detection; edge computing; multi-modal fusion; graded response; automotive-grade MCU**

I. Introduction

Fatigued driving is a leading cause of traffic accidents worldwide, accounting for twenty percent to thirty percent of fatal crashes and over forty percent of major traffic accidents [1]. Traditional driver fatigue detection schemes based on X86 or ARM Cortex-A architectures suffer from high costs and excessive computing power requirements, limiting their popularization in entry-level vehicles and low-cost automotive electronic systems. With its open-source nature, low power consumption, and emerging automotive-grade certifications, the RISC-V architecture is gradually entering the field of automotive electronics[2]. RISC-V is an open-source and royalty-free instruction set architecture (ISA) with the characteristics of modularization and extensibility, explicitly supporting domain-specific custom extensions [3]. It aims at becoming a universal ISA by supporting numerous processor sizes, from embedded controllers to high-performance computers, and a wide variety of software stacks and programming languages [4]. However, there remains a lack of lightweight fatigue detection solutions optimized for and demonstrated on RISC-V processors.

Existing fatigue detection methods are generally categorized into three types: physiological signal-based, vehicle behavior-based, and machine vision-based [5]. Physiological signal-based methods [6,7], such as those using EEG or ECG, offer high accuracy but require contact sensors that can interfere with driving. Vehicle behavior-based methods [8-10], which monitor steering wheel operations or lane deviations, are non-intrusive but are affected by individual driving habits and road conditions, leading to poor robustness. Machine vision-based methods [11, 12], which analyze facial features, are non-intrusive and easy to implement but often struggle with complex lighting conditions and low hardware performance constraints.

In the field of fatigue driving detection, researchers have proposed various improved algorithms. For example, a fatigue driving detection algorithm based on facial multi-feature fusion combining driver characteristics introduces an improved YOLOv3-tiny convolutional neural network to capture facial regions under complex driving conditions, and constructs driver identity information library and fatigue assessment model to improve detection accuracy [13]. Another method combines BP neural network and time cumulative effect, using Adaboost algorithm for face detection, Kalman filter for face tracking, and cascade regression tree for facial landmark detection, which effectively filters out false detections caused by facial expressions [14]. These research results provide valuable references for the optimization of fatigue detection algorithms.

To address these issues, this paper proposes a lightweight edge computing-main control architecture for RISC-V. Compute-intensive image processing tasks are offloaded to a K210 vision module, while the main RISC-V controller focuses on message parsing and fusion decision-making. The system integrates visual-pressure dual-modal detection, visual-auditory graded alarms, and hardware modularization through standardized interfaces.

The main contributions of this work are threefold:

- A Layered RISC-V System Architecture for Low-Cost Scenarios: By completely offloading computationally

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intensive vision algorithms to an independent edge perception module (K210), the main controller (RISC-V MCU) is only responsible for lightweight logic decisions and peripheral control. This design significantly reduces the computational demand on the main control unit, enabling the use of a low-cost automotive-grade 32-bit RISC-V MCU and providing a new approach to overcome the cost barriers of traditional high-performance processor-dependent solutions.

- Engineering Integration of a Lightweight Dual-Modal Detection Strategy: On resource-constrained edge platforms, classic multi-feature visual detection algorithms (EAR/PERCLOS/MAR) were adapted and optimized. By integrating steering wheel grip pressure sensing, a decision fusion mechanism centered on vision with pressure as auxiliary verification was constructed. This strategy ensures low computational occupancy while improving detection robustness, particularly in ambiguous ”fatigued” states.

- A Replicable Prototype Reference for RISC-V Automotive Safety Systems: A three-level graded response logic and standardized communication interfaces were designed and implemented, completing a full-link laboratory validation from perception to decision-making to execution. The modular design facilitates easy porting to different RISC-V hardware platforms, offering a concrete engineering example for the application of RISC-V in entry-level automotive electronics.

II. Overall System Design

A. Design Objectives

The core objectives of the system include low cost, high portability, and strong real-time performance. The specific technical indicators are listed as follows:

- The hardware cost is no more than five hundred RMB.
- Three-level fatigue detection (normal, fatigued, severely fatigued) with an overall accuracy of no less than eighty-five percent, reaching no less than ninety-two percent under optimal conditions, and an alarm latency of less than three hundred milliseconds.
- Support for multi-modal information fusion and rapid porting to other RISC-V platforms.
- Low-power consumption and stable operation in vehicle-mounted environments ranging from eight volts to thirty-five volts, with basic hardware protection functions.

B. System Architecture

The system adopts a three-level architecture of perception layer, control layer, and execution layer based on edge computing:

- Perception layer: Consists of a K210 visual unit and an STM32 pressure unit. The K210 unit drives an OV2640 camera for QVGA image collection, performing face detection, eye key point extraction, and multi-feature (EAR/PERCLOS/MAR) fusion calculation to generate 1-byte fatigue state flags (0/1/2). The STM32 unit collects steering wheel grip pressure signals via an ADC interface, generating 1-byte pressure state flags (0/1). Both units transmit data to the main controller via UART.
- Control layer: The E-Silicon EBP3901 development board (equipped with the EAM2011 MCU) implements UART data parsing using an interrupt-driven mechanism, performs dual-modal (visual-pressure) fusion judgment through a three-level state machine, and outputs execution instructions to the execution layer.
- Execution layer: Utilizes an OLED display and an active buzzer to implement visual and auditory graded alarms.

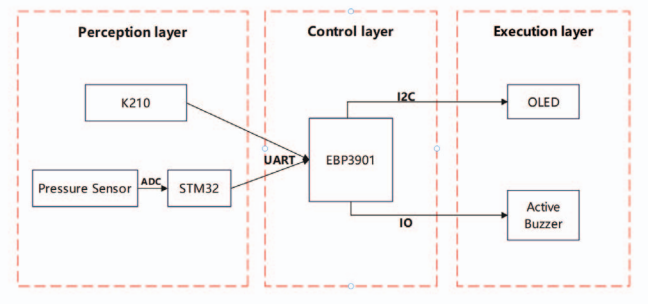


Fig. 1. Overall System Architecture Diagram

C. Hardware Selection and Considerations

All hardware components are selected for low cost and standardization:

- Main control: EBP3901 development board (EAM2011 MCU). While this prototype uses a development board, the design is intended for integration with an automotive-grade MCU featuring standard interfaces and low power consumption.
- Visual sensing: K210 open-source camera module, chosen for its hardware acceleration of lightweight computer vision algorithms, enabling efficient execution of face detection.
- Auxiliary sensing: STM32F103C8T6 MCU + FSR408 thin-film pressure sensor, mounted on the steering wheel. The use of a separate MCU for pressure sensing allows for modular development and reduces the processing load on the main RISC-V controller.
- Execution/Alarm: I2C OLED display and GPIO-controlled active buzzer.

Regarding the three-chip architecture: The ”K210 + STM32 + RISC-V” design is a deliberate choice for modularity and engineering scalability. It decouples perception (vision, pressure) from decision-making (RISC-V). This allows the main controller to be flexibly replaced with different RISC-V chips or even MCUs of other architectures without altering the perception algorithms, greatly enhancing the solution’s portability. It also enables independent upgrades of the vision algorithm or the addition of new perception modalities (e.g.,

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acoustic) without impacting the core control logic, aligning with the trend towards modularization in automotive electronics. RISC-V's modular design enables optional standard extensions that the hardware can include if specified by the system's requirements, facilitating the development of small or large-scale applications . RISC-V explicitly supports domain-specific custom extensions, which can be tailored to the specific needs of automotive electronic systems .

III. Key Technology Implementation

A. Layered Design of Software Architecture

The software architecture adopts a layered design, with independent development for the perception layer and control layer, communicating via a lightweight protocol:

- K210 Software: Built on the CanMV framework, this module implements face detection via the YOLOv2 algorithm, extracts 68 facial key points, and computes the Eye Aspect Ratio (EAR), Percentage of Eyelid Closure Over the Pupil Over Time (PERCLOS), and Mouth Aspect Ratio (MAR). A 60-second sliding window combined with first-order low-pass filtering is applied to smooth the Beats Per Minute (BPM) values. Fatigue states are digitally encoded (0: normal, 1: fatigued, 2: severely fatigued) and transmitted via Universal Asynchronous Receiver/Transmitter (UART).
- STM32 Software: Developed with Keil5 IDE, this module collects pressure data through Analog-to-Digital Converter (ADC), and calculates statistical features including mean, maximum, minimum, and variance. Pressure anomalies persisting for no less than ten seconds are flagged as abnormal (encoded as 1), while the recovery to normal status lasting for no less than five seconds is flagged as normal (encoded as 0). All data is transmitted via UART.
- RISC-V Main Control Software: Based on Eswin IDE, this module parses UART data through interrupt-driven mechanisms, executes the multi-modal fusion algorithm, and drives peripheral devices. RISC-V supports three privilege levels of execution to protect different software stacks, enabling multiple software stacks to run with different privileges levels . This feature ensures the security and stability of the main control software. RISC-V's software ecosystem includes compilers, state-of-the-art OSs, and system generation tools to ease hardware/software interfacing, which provides good support for the development of main control software .

B. Equations

A lightweight multi-feature fusion algorithm is implemented on the K210:

- The eye aspect ratio (EAR) is defined as twice the sum of the vertical eye distances divided by the horizontal eye distance, where d1 and d2 denote the vertical eye distances, and d3 denotes the horizontal eye distance,hysteresis thresholds (0.28 for closure, 0.32 for opening) and a 60-ms refractory period are adopted to filter jitter.

$$EAR = 2(d1 + d2)/d3 \tag{1}$$

- The percentage of eyelid closure over the pupil over time (PERCLOS) is defined as the ratio of the number of frames with EAR below 0.28 to the total number of frames within a 300-frame detection period,N1 represents the number of frames with EAR¡0.28, and N2 =300 denotes the total number of frames in the detection interval. A threshold value of 0.21 is employed to determine the fatigue state.

$$PERCLOS = N1/N2 \tag{2}$$

- The mouth aspect ratio (MAR) is calculated as the ratio of the sum of vertical mouth distances to twice the horizontal mouth distance,where d4 and d5 denote the vertical mouth distances, and d 6 denotes the horizontal mouth distance. A threshold of 0.6 is adopted for MAR to detect yawning.

$$MAR = (d4 + d5)/(2d6)labeleq \tag{3}$$

The visual state judgment is performed based on the following criteria: the state is judged as fatigued if PERCLOS is greater than twenty-one percent or MAR is greater than zero point six; it is judged as severely fatigued if both of the above conditions are met, or the BPM ratio is less than zero point six; otherwise, the state is judged as normal.(The thresholds (EAR=0.28, PERCLOS=0.21, MAR=0.6) are not arbitrary but were calibrated during preliminary experiments based on the specific hardware configuration (OV2640, QVGA@15fps). Data from 10 healthy subjects (22-35 years old) in a simulated driving environment were used to establish baseline values that ensure stable system operation under our fixed experimental conditions.)

C. Dual-Modal Fusion Strategy

A "vision-primary, pressure-auxiliary" fusion strategy is adopted. This design prioritizes low false positive rates, a core requirement for user acceptance in entry-level vehicles. Pressure signals, being susceptible to driving maneuvers, are used to reinforce decisions only in the ambiguous "fatigued" state.

When the visual state is normal (0), the overall state is judged as non-fatigue, regardless of pressure. This prevents false alarms from normal steering variations.

When the visual state is fatigued (1), if the pressure state is abnormal (1), the overall state is upgraded to severely fatigued to reduce missed detections; otherwise, it remains fatigued.

When the visual state is severely fatigued (2), the result remains unchanged regardless of the pressure state, as the visual evidence is already conclusive.

D. Graded Response and Peripheral Control

A three-level graded response mechanism is implemented: Normal (0): OLED shows "Normal Driving", BPM value;

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buzzer off.
Fatigued (1): OLED shows ”Fatigued! Please take a break”; buzzer emits intermittent alarm (on 10ms/off 70ms).
Severe Fatigue (2): OLED shows ”Severely Fatigued! Stop driving immediately”; buzzer emits rapid alarm (on 5ms/off 20ms).
Non-blocking drivers are used for the buzzer and line-refresh for the OLED to minimize resource occupation and ensure real-time performance.

E. Communication Protocol Design and Reliability Considerations

A single-byte minimal UART protocol (baud rate 115200) is designed for low latency in the prototype:
K210 transmits 0x00, 0x01, or 0x02 based on visual state.
STM32 transmits 0x00 or 0x01 based on pressure state.
The main controller receives data in interrupt-driven mode with a buffer.

Note on Reliability: This simple protocol is intended for functional verification in a controlled lab environment. It lacks checksums and retransmission mechanisms and is not suitable for the high electromagnetic interference (EMI) of a real vehicle. For industrial-grade deployment, this protocol would be enhanced with standard reliability features like CRC checks, frame headers, and timeout retransmission mechanisms.

IV. System Testing and Results

A. Development Environment

- Hardware:EBP3901 dev board,K210, STM32F103C8T6,OV2640, OLED,buzzer,FSR408.
- Software: Windows 11, Eswin IDE, CanMV IDE, Keil5.

B. Test Methods and Indicators

Performance tests were conducted in a simulated driving environment. The ground truth for fatigue was based on the Karolinska Sleepiness Scale (KSS): scores 1-3 as ”normal”, 4-6 as ”fatigued”, and 7-9 as ”severely fatigued”. Ten healthy subjects (age 22-35, 5 male, 5 female, all with driving licenses) participated. Each subject was tested under three lighting conditions (strong: 1000 lux, soft: 500 lux, weak: 100 lux). Each test condition was repeated three times, with each session lasting 20 minutes, and results were averaged. A total of 10 (subjects) × 3 (lighting) × 3 (repetitions) = 90 sessions were conducted. Data from the final 10 minutes of each session were used for analysis to ensure the subject had settled into a steady state. A total of 54,000 image frames and corresponding pressure data were analyzed.

C. Test Results and Analysis

As shown in Table 1, under soft light conditions, the system achieves an overall detection accuracy of ninety-two point five percent, which satisfies the design objective of no less than ninety-two percent under optimal conditions. The average alarm delay is measured to be less than two hundred and fifty milliseconds, which also meets the requirement of less than three hundred milliseconds. The communication bit error rate in the laboratory environment is less than zero point zero one percent.

TABLE I
System Detection Accuracy under Soft Light (500 lux)

Fatigue State (KSS)	Samples (Frames)	Correct Detections	Detection Accuracy
Normal (KSS 1-3)	18,000	16,704	0.928
Fatigued (KSS 4-6)	18,000	16,650	0.925
Severely Fatigued (KSS 7-9)	18,000	16,596	0.922
Overall	54,000	49,950	0.925

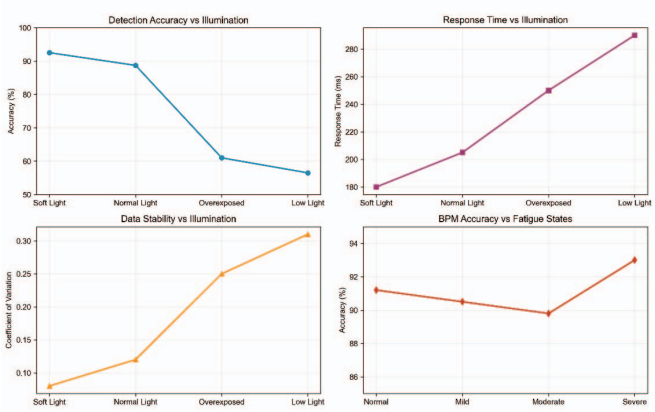


Fig. 2. System Detection Accuracy under Different Lighting Conditions

As shown in Figure 2, system performance degrades under non-optimal lighting conditions. Accuracy drops to eighty-six point three percent under strong light of one thousand lux and eighty-four point one percent under weak light of one hundred lux. This indicates that while the system performs well in controlled environments, its sensitivity to extreme lighting constitutes a limitation, as noted in the design objectives. This is primarily due to the limited dynamic range of the OV2640 sensor and the absence of adaptive algorithms in the current prototype.

TABLE II
Comparison with Existing Low-Cost Visual Detection Schemes

Algorithm	Core Technology	Reported Accuracy	Approx. Hardware Cost
Proposed in [11]	Facial landmark + SVM	0.943	100 dollar
Proposed in [12]	Iris tracking + PERCLOS	0.808	50 dollar
This Work	Multi-feature fusion + Pressure	0.925	70 dollar

Compared to existing low-cost visual-only schemes, this work achieves competitive accuracy while adding a pressure modality for enhanced robustness and maintaining a low overall system cost.

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TABLE III
HARDWARE COST BREAKDOWN

Component	Model	Unit Price (RMB)
Main Control Board	EBP3901 (EAM2011)	180
Vision Module	K210 + OV2640	150
Pressure Sensing Module	STM32F103 + FSR408	120
Display	OLED 0.96 inch	20
Buzzer and other passives	-	30
Total	-	500

D. Scheme Improvement Strategies

To address the identified limitations, the following strategies are proposed for future work:

- Enhanced Environmental Adaptability: Introduce a near-infrared (NIR) camera and active NIR illumination to make the system invariant to visible light changes. This is a more robust solution than a simple fill light.
- Richer Detection Dimensions: Integrate an IMU sensor to detect head nodding and posture, as explored in vehicle behavior-based methods [4]. This can serve as an additional validation cue for severe fatigue.
- Refined Multi-Modal Fusion: Evolve the current rule-based fusion to a more sophisticated model. For instance, features from the pressure signal (e.g., mean, variance over a sliding window, as used in [6]) could be extracted and fed into a lightweight classifier (e.g., a small neural network) on the RISC-V MCU, allowing for more nuanced fusion than a simple binary flag.
- Adaptive and Lightweight Algorithms: Replace the current YOLOv2 with a more efficient architecture, such as the improved YOLOv11n with an EfficientNetV2 backbone mentioned in [2], which reportedly reduces parameters by fifty percent while improving accuracy. This would free up resources on the K210 for more complex preprocessing or higher frame rates. Implement an adaptive algorithm for calibrating EAR/PERCLOS thresholds to individual drivers during the first few minutes of use, reducing inter-subject variability.

V. CONCLUSION

This paper designs and implements a RISC-V-based driver fatigue detection and graded response system prototype. By leveraging a layered edge computing architecture, multi-feature visual detection, and a ”vision-primary, pressure-auxiliary” fusion strategy, the system addresses the high cost and high computational demands of traditional schemes. With a total hardware cost of under five hundred RMB, the system achieves a face-eye detection accuracy of no less than ninety-two point five percent under soft light and an alarm delay of less than three hundred milliseconds in a simulated environment, successfully validating the proposed architecture.

However, this study has clear limitations. The current system’s performance degrades under extreme lighting, its dual-modal fusion strategy is simplistic, its thresholds are not yet adaptive, and the prototype lacks the robustness needed for real automotive applications such as electromagnetic interference protection and power supply protection.

Future work will focus on:
(1) Developing more robust vision algorithms, potentially using near-infrared imaging, to handle complex lighting conditions.
(2) Designing adaptive threshold calibration and a more refined, feature-based multi-modal fusion algorithm. Utilizing RISC-V’s custom extension capability to optimize the fusion algorithm .
(3) Strengthening the hardware circuit with proper power supply protection and electromagnetic interference countermeasures according to automotive standards.
(4) Conducting long-term on-road validation tests.

This work provides a foundational reference for the deployment of RISC-V in the entry-level automotive safety market.

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